

**BTECH
(SEM V) THEORY EXAMINATION 2024-25
DATA ANALYTICS**

TIME: 3 HRS

M.MARKS: 70

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A

1. Attempt all questions in brief.

2 x 07 = 14

Q no.	Question	CO	Level
a.	Differentiate between Predictive and Prescriptive Data Analytics.	1	K2
b.	Define the term data lake, data base and data warehouse.	1	K1
c.	Explain the concept of Outliers.	2	K2
d.	Describe the concept of Lasso Regression.	2	K2
e.	Differentiate between Steam Processing and Traditional Data Processing.	3	K2
f.	Write the two limitations of K-Mean.	4	K1
g.	Discuss the various categories of clustering techniques.	5	K2

SECTION B

2. Attempt any three of the following:

07 x 3 = 21

a.	Explain the different categories of data analytics with examples.	1	K2
b.	Explore PCA. Given data = {4, 8, 13, 7; 11, 4, 5, 14}. Compute the principal component using PCA algorithm. Also use PCA to reduce dimension from 2 to 1.	2	K3
c.	Explore the term- Market Basket Analysis. Is it supervised or unsupervised? Determine how would a company use market basket analysis to improve its marketing strategies?	3	K3
d.	Differentiate between CLIQUE and ProCLUS clustering	4	K4
e.	Differentiate between NoSQL database and a Relational database. Identify when one should use a NoSQL database instead of a relational database with a suitable example.	5	K4

SECTION C

3. Attempt any one part of the following:

07 x 1 = 07

a.	Differentiate between Structured data, Semi-structured data and Unstructured Data.	1	K2
b.	Describe Big Data and its characteristics.	1	K2

4. Attempt any one part of the following:

07 x 1 = 07

a.	Differentiate between Neural Network and Artificial Neural Network.	2	K2
b.	A = {(10,0.2), (20,0.4), (25,0.7), (30,0.9), (40,1), (50,0.4)} B = {(10,0.4), (20,0.1), (25,0.9), (30,0.2), (40,0.6), (50, 0.6)} Apply Union, Intersection, Complement, Bold Union and Bold Intersection operations on above listed Fuzzy Sets.	2	K3

5. Attempt any one part of the following:

07 x 1 = 07

a.	Explain and apply Flajolet-Martin algorithm on the following stream of data to identify unique elements in the stream. S=1,3,2,1,2,3,4,3,1,2,3,1 Given: $h(x)=(6x+1) \bmod 5$	3	K3
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Subject Code: BCS052

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b.	Discuss the Concept of filtering in Data Stream Processing. Explain Bloom Filtering in detail.	3	K2
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6. Attempt any one part of the following: 07 x 1 = 07

a.	Cluster the following eight points (with (x, y) representing locations) into three clusters: A1(2, 10), A2(2, 5), A3(8, 4), A4(5, 8), A5(7, 5), A6(6, 4), A7(1, 2), A8(4, 9). Initial cluster centers are A1(2, 10), A4(5, 8) and A7(1, 2). The distance function between two points a = (x1, y1) and b = (x2, y2) is defined as $P(a, b) = x2 - x1 + y2 - y1 $ Use K-Means Algorithm to find the three cluster centers after implanting all eight points.	4	K3														
b.	The database has 6 transactions. Assume Support threshold=50%, Confidence= 60% <table border="1"><thead><tr><th>TID</th><th>Items Bought</th></tr></thead><tbody><tr><td>10</td><td>Beer, Nuts, Diaper</td></tr><tr><td>20</td><td>Beer, Coffee, Diaper</td></tr><tr><td>30</td><td>Beer, Diaper, Eggs</td></tr><tr><td>40</td><td>Nuts, Eggs, Milk</td></tr><tr><td>50</td><td>Nuts, Coffee, Diaper, Eggs, Milk</td></tr><tr><td>60</td><td>Beer, Nuts, Diaper</td></tr></tbody></table> i) Use Apriori algorithm to find all frequent itemsets. ii) Show all the strong association rules (with support and confidence)	TID	Items Bought	10	Beer, Nuts, Diaper	20	Beer, Coffee, Diaper	30	Beer, Diaper, Eggs	40	Nuts, Eggs, Milk	50	Nuts, Coffee, Diaper, Eggs, Milk	60	Beer, Nuts, Diaper	4	K3
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7. Attempt any one part of the following: 07 x 1 = 07

a.	Brief about the main components of MapReduce.	5	K2
b.	Draw the architecture of HIVE with its features.	5	K2